

Understanding of sliding and contact angle results in tilted plate experiments

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Abstract

Tilted plate experimental methodology is studied in order to address misinterpretations and omissions that exist in the literature. The effect of methodology on subsequent sliding angle measurements is quantitatively assessed in two parts. The first part is a comparison of drop placement techniques in the conventional tilted plate method (drops placed on a level surface, then inclined). The second part is a comparison between the conventional tilted plate method and a modified method, in which the drop is placed on a surface that is already inclined. The first part of the study involved water drops placed on alkyl ketene dimmer (AKD) surfaces using a needle from above as well as below the surface (through a hole). For drops placed from below, three different drop locations (with respect to the hole in the surface) are included in the study. It was found that the drop placement technique had a statistically significant impact on sliding angle measurements, and the most consistent and unbiased measurements resulted when the drop was placed from below, with the hole leading (downhill from) the drop. The second part of the study involved water drops on both AKD (high contact angle hysteresis system) and fluorinated silicon (low contact angle hysteresis) surfaces. For either system, it was shown that sliding angles were lower when drops were placed on surfaces that were already inclined. Since the maximum and minimum contact angles were identical between the two experimental methods, the difference in adhesion was due to variation in contact line shape. The results of this study show that errors of 50–60% would result from using conventional tilted plate measurements to predict droplet mobility on fixed, inclined surfaces, such as those found in condensers. The important distinction between repellency (measured by the advancing contact angle) and drop mobility is also discussed. Further, it was observed that the advancing and receding contact angles do not always correspond to the maximum and minimum contact angles observed in a tilted plate experiment.

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1. Introduction

In a typical wetting experiment using the tilted plate method, a drop is placed onto a surface, which is then inclined until the drop begins to slide. The sliding angle is a measure of the mobility of a drop on the surface, which is of concern to a variety of applications from textiles to microfluidics. The tilted plate method is also sometimes used to determine the advancing and receding contact angles. However, this may not be the most suitable method, as will be discussed. Unfortunately, studies that use the tilted plate method are seldom accompanied by methodological details much beyond providing an apparatus model number. It will be seen that a number of important subjectivities are thereby

withheld: from the drop placement technique to the precise definition of sliding angle (the latter being especially important for verification of theoretical results). These issues not only preclude surface performance comparisons between different researchers but can obscure the physical meaning of the results.

An example of a misinterpretation arising in the literature is the relationship between drop mobility and repellency. Repellency is a measure of a surface's resistance to "wetting out." Repellency can be defined by the thermodynamic quantity of the equilibrium contact angle, but due to the impracticality of equilibrium contact angle measurements, it is typically represented by the advancing contact angle of a drop, which is measured on a level surface. Öner and McCarthy [1] have prudently drawn attention to recent literature that indiscriminately associates advancing contact angle with mobility. However, they go on to stress the importance of contact angle hysteresis to mobility, citing the original equation, derived by Fumridge [2],

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that defined sliding angle as a function of the advancing and receding contact angles (θ_A and θ_R):

$$\sin(\alpha) = \gamma \frac{Rk}{mg} (\cos \theta_R - \cos \theta_A) \quad (1)$$

where α is the sliding angle, γ is surface tension, m is the mass of the drop, and R and k are a length scale and shape constant for the contour of the drop, respectively (R is generally taken as the drop radius and k is a fitting parameter based on the experimental data). Although Öner and McCarthy made it clear that contact angle hysteresis can be a qualitative indication of drop mobility, it has been argued by Krasovitski and Marmur [3] that advancing and receding contact angles, as measured on a level surface, should theoretically not be used in numerical predictions of sliding angles (a measure of drop mobility). Instead, the authors define the maximum and minimum contact angles ($\theta_{\max}$ and $\theta_{\min}$), which are those at the leading and trailing edges of a drop profile on a surface inclined to the sliding angle (Fig. 1a). They report that theoretical evidence suggests that the relationship between $\theta_{\max}$ and θ_A as well as $\theta_{\min}$ and θ_R varies between different surface/liquid combinations. Although this conclusion disagrees with the empirical data of ElSherbini and Jacobi [4], which exhibit $\theta_{\max}$ and $\theta_{\min}$ approximately equal to θ_A and θ_R for all surface/liquid combinations, the results of this study support it. The contact angles of water drops on both alkyl ketene

dimmer (AKD) and fluorinated silicon surfaces were measured both at the sliding angle as well as on a level surface as water was added and withdrawn from the drop. The receding contact angle (θ_R) of a water drop on AKD surface was at least 73° less than the minimum contact angle ($\theta_{\min}$), but for fluorinated silicon surfaces, they were not measurably different. Therefore, the relationships between the advancing/receding and maximum/minimum contact angles are unique consequences of surface/liquid combinations, so Eq. (1) must be redefined as follows:

$$\sin(\alpha) = \gamma \frac{Rk}{mg} (\cos \theta_{\min} - \cos \theta_{\max}) \quad (2)$$

As in Eq. (1), R and k are length scale and shape constants for the contour of the drop. These parameters are necessary because the sliding drop problem cannot be adequately represented by a two dimensional profile image of the drop (providing $\theta_{\max}$ and $\theta_{\min}$). Drop adhesion is the result of a distributed force acting parallel to the surface and perpendicular to the contact line (Fig. 1b). The magnitude of the force distribution is determined by the contact angle at each point, but its direction results from the shape of the contact line (represented by the fitting parameter k). Therefore, the shape of the contact line defines the component of the force distribution that is active in adhesion, i.e. opposes the component of gravitational force acting along the surface plane. It will be seen that the contact line shape has a significant influence on mobility. As such, drop mobility is not a primary thermodynamic property, as it is the manifestation of the interaction between an external force and drop adhesion force.

Precision is an issue of great concern with tilted plate experiments. To obtain statistically significant results, it is often necessary to repeat experiments many times. Furthermore, it has been observed that small differences in experimental techniques amongst researchers can dramatically affect the sliding angle. In order to increase measurement confidence as well as facilitate the comparison of data between researchers, detailed tilted plate experiment protocol must be discussed. This study examines several points regarding tilted plate experiment protocol with respect to their affect on the magnitude and precision of the results. In particular, it will be shown that differing placement methods of a water drop on AKD can significantly affect the magnitude and precision of sliding angle measurements.

Tilted plate methodology must be considered not only in terms of experimental precision, but also in terms of its applicability to the physical problem in question. Certainly, not all fluid shedding applications emulate a conventional tilted plate experiment, wherein drops are placed on level surfaces that are subsequently inclined. For example, consider a windshield, which is already inclined when drops land on it. Can we predict mobility behavior for all applications based on the results of conventional tilted plate experiments?

To illustrate the importance of this question, the problem of drop growth and shedding on fixed, inclined surfaces, such

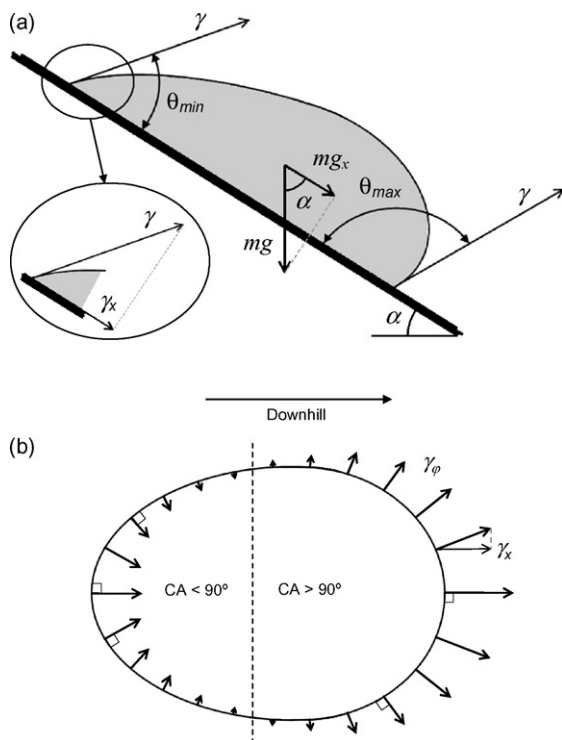


Fig. 1. (a) Illustration of the maximum ($\theta_{\max}$) and minimum contact angles ($\theta_{\min}$) as well as the body gravity force (mg), and surface tension force (γ) for a drop inclined to the sliding angle (α); the components of these forces, γ_x and mg_x , act parallel to the surface and perpendicular to the axis of inclination. (b) Top view of the component of the surface tension force distribution that is parallel to the surface (γ_ϕ) for a drop on an incline; dashed line shows contact angle transition through 90° . Adhesion (reaction) forces are equal and opposite to the component of (γ_ϕ) that is perpendicular to the axis of inclination (γ_x).

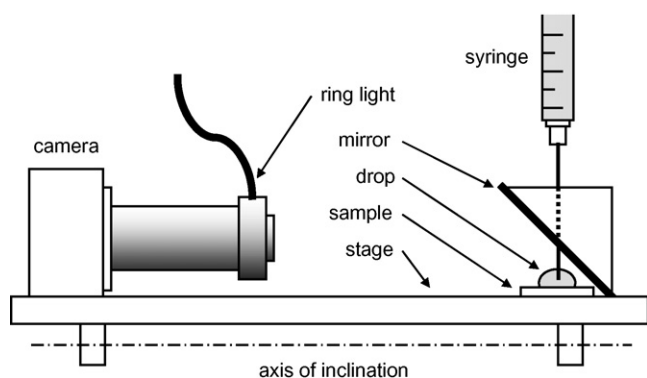


Fig. 2. Illustration of 45° mirror block used to obtain top view of drops with camera.

as in a condenser, is investigated. It will be shown that the contact line shape, and subsequent sliding angle measurements, can change depending on whether the drop is placed using the conventional method, on a level surface that is subsequently inclined, or on a surface that is already inclined. Callaghan and Amirfazli [5] have reported a discrepancy in sliding angles between the two methods and linked this behavior to a marked difference in the minimum contact angle, rather than contact line shape. However, it was unclear whether the results of the study had been affected by the addition of kinetic energy when the drop was placed by hand on the surface. In the current study, drop placement/growth has been automated and slowed in order to reduce kinetic energy input.

2. Apparatus

Profile images of drops were taken using a CCD camera (A302f, Basler AG) at 640×480 pixels; drops were backlit with a diffused 10 W halogen lamp. The camera was mounted to a custom stage that was inclined by a linear motor (P01-23 $\times$ 80/150 $\times$ 210, LinMot) at an angular speed of $1.2^\circ/\text{s}$ and acceleration of $1.0^\circ/\text{s}^2$. Vibration from the linear motor was isolated from the stage using rubber grommets. There were two stepper servo (HT11-012, Applied Motion Products) controlled syringes (#1725 GasTight, Hamilton) with 26G needles, which could place a drop from either above or below the surface (through a hole drilled in the sample). The inclination angle of the stage was measured using a two-axis inclinometer (TSD-90, Instruments & Control Inc.), which was calibrated to be accurate within $\pm 0.01^\circ$ from 0 to 60° . Each experiment was entirely software controlled and automated to ensure consistency.

Top views of the drops are reflections in a mirror positioned above the drop, at 45° to the camera view plane. Lighting was provided by an incident ring light that was mounted on the camera lens. Fig. 2 is a diagram of this arrangement.

Evaporation of the drop was measured and deemed to have an insignificant effect on the results of the study. However, different study durations, surface materials, and test liquids may warrant more attention to this problem. One possible solution is a clear enclosure that is filled with air saturated with the test liquid vapor [6].

3. Methodology

3.1. Sample preparation

This study involved two types of hydrophobic surfaces, which lie on either end of the roughness spectrum: textured alkyl ketene dimmer (AKD) and fluorinated (smooth) silicon wafer. The AKD surfaces were superhydrophobic and were produced as per the directions of Mohammadi et al. [7]. AKD pellets (Aqualap 364, Hercules Inc.) were melted and heated to 90°C , then immediately allowed to cool down to approximately 35°C , when the molten pellets began to nucleate. The AKD was then poured into a stainless steel mold with aluminum heat sinks, which, after a few minutes, was opened and placed in a dry nitrogen environment for 72 h. During this time, the open face of the AKD surface formed a random microstructure (texture) with an RMS roughness of $2 \pm 1 \mu\text{m}$ (AFM was used to find the RMS roughness value, note that the maximum roughness was one order of magnitude larger). The resulting 2 mm thick surface was then sectioned into approximately 15 mm square samples using a hot knife and stored with a desiccant. If needed, a hole was drilled through the sample by hand, using a beveled-tip syringe needle.

Fluorinated silicon samples were produced from polished wafers (Type P/Boron Orient 100, University of Alberta Nanofab) that were diced into approximately 15 mm^2 samples, soaked overnight in chromic–sulfuric acid, and then thoroughly rinsed with distilled water. A 6% Teflon solution (Teflon AF 1600, DuPont Co.) was diluted in a solvent (FC-75, DuPont Co.) at a volumetric ratio of 1:3. Approximately $100 \mu\text{l}$ of this solution was applied to each sample and leveled in a spin coater (Model 6700, Specialty Coating Systems Inc.). The coated samples were allowed to dry overnight under vacuum. If needed, a hole was drilled through the sample using a 0.46 mm diamond drill (MCDU18, UKAM Industrial Superhard Tools). The roughness of surfaces produced in this way is generally in the order of 1 nm.

3.2. Advancing and receding contact angle measurement

The advancing and receding contact angles were obtained by adding and withdrawing water from a drop on a level surface. Although it is common practice to manipulate the drop volume with a needle that is immersed into the drop from above, it is preferred to inject liquid from below, through a hole, in order to avoid possible influence of the needle on the observed contact angles, especially for small drops. The samples were drilled and placed on the stage so that the needle ran through the hole, with its tip just below the surface of the sample. Water was added and withdrawn from the drop very slowly ($0.2 \mu\text{l}/\text{s}$) to avoid dynamic effects, and an image of the drop profile was taken every 4 s. In three separate experiments, water was added to the drops until they reached maximum volumes of 30, 50, and $70 \mu\text{l}$, then water began to be withdrawn. These volumes correspond to contact line diameters of approximately 4, 5, and 6 mm for water on fluorinated silicon surface and 2, 3, and 4 mm on AKD surface.

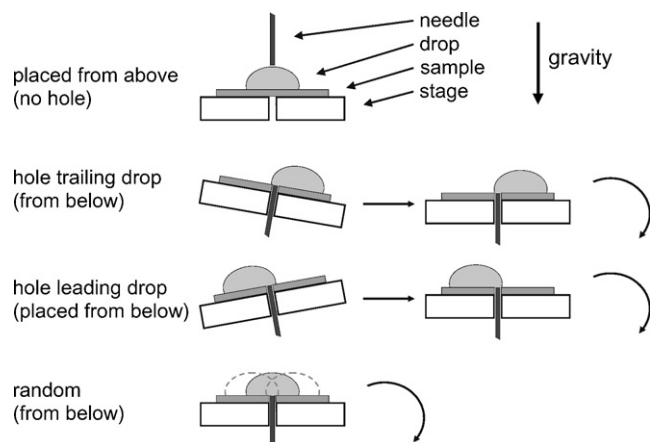


Fig. 3. Four-drop placement methods for conventional tilted plate experiments. To place a drop with the hole trailing or leading it, the surface is inclined 3° forward or backward, respectively, while the drop is formed on the surface. Before the experiment, in which the surface is tilted slowly forward, the surface is always brought back to level.

3.3. Drop placement in conventional tilted plate experiments

Drops of water were placed on AKD surface using four practical methods in order to ascertain their effect on the sliding angle or experimental precision. The four placement methods are illustrated in Fig. 3: from above (no hole on surface); from below, hole leading the drop; from below, hole trailing the drop; and from below, random position of drop with respect to the hole. In order to place the drop so that the hole was leading or trailing the drop, the stage was tilted 3° backwards or forwards, respectively, then the drop was formed; the surface was subsequently leveled before the start of the experiment (see Fig. 3). In the random placement method, the surface was leveled, and then the drop was formed from below; the position of the drop with respect to the hole was not controlled. In placing the drop from above, a $10\ \mu\text{l}$ drop was transferred from the needle tip to a level surface, then water was added to the drop at a rate of $1\ \mu\text{l/s}$ with the needle tip positioned just above the surface of the drop, without ever touching it. The needle tip could not be immersed in the drop because the contact angle would have receded slightly when the needle was retracted. For repeatable sliding behavior, the drop must be placed in an advancing contact angle regime each time. When an axisymmetric drop is advancing, it is insured that the contact line length is identical for a given drop volume.

After placing the drop on the surface using any of the above four techniques, the surface was inclined in increments of approximately 2° . At each increment, an image of the drop profile was saved before and after a 5 s pause. The purpose of this pause was to observe whether the drop would move without the kinetic input of the inclining stage. The before and after images could later be compared to determine whether the entire drop had slipped down the surface, as indicated by a discernible movement of both the leading and trailing edges *without* any change in the contact angles. A change in contact angle would indicate that only a portion of the contact line had slipped, an otherwise indiscernible event in a profile view of the drop. For

each drop placement method, this procedure was repeated eight times, each time using a fresh AKD surface sample. The volume of the drop was always $30\ \mu\text{l}$.

The definition of the sliding angle is not trivial. A drop seldom slides completely off a surface at a given inclination angle, but instead tends to slide at a speed that is related to the angle of inclination (i.e. higher inclination, higher sliding speed). Therefore, the length of the pause determines the sliding speed (and corresponding inclination angle) necessary for a discernible movement between the before and after images. A 5 s pause was a subjective choice that was judged to detect movement slight enough to be considered incipient motion while not being excessively time-consuming.

3.4. Handheld drop placement on inclined surfaces

The experiments of Callaghan and Amirfazli [5] were repeated with AKD and fluorinated silicon surfaces. Drops of water were placed on the surfaces using a handheld digital pipette (Nichipet EX, Fisher Scientific). The sliding angles of two experimental methods were compared: placing a drop on a level surface, then inclining it, and placing a drop on a surface that is already inclined. In the latter method, the surface was inclined at some angle, and a $20\ \mu\text{l}$ drop was placed on the surface. If this drop did not slide, then the volume of the drop was increased in increments of $10\ \mu\text{l}$ until it did. Because drop placement was not automated, the point of sliding had to be determined with the naked eye, rather than as a comparison between consecutive images. For this reason, the sliding angle measurements for handheld drop placement were slightly higher in magnitude and lower in precision than those for the automated drop placement method. Although direct numerical comparison between these sliding angle measurements and those in the rest of this study is not recommended, their trends remain comparable. All experiments were repeated between three and four times for each data point, which was sufficient to establish a consistent trend.

3.5. Automated drop placement on inclined surfaces

In order to avoid kinetic energy input (e.g. dropping, slight pushing, or vibration) from the handheld pipette, an automated method of drop placement was conceived. On a level surface, a “seed” drop was grown from above at a rate of $1\ \mu\text{l/s}$, and the needle tip of a servo operated syringe was immersed into its center. The volume of the seed drop was chosen to be $10\text{--}30\ \mu\text{l}$ less than the volume at which the drop was expected to slide (e.g. a $40\text{--}60\ \mu\text{l}$ seed if the drop was expected to slide at $70\ \mu\text{l}$). The surface was then inclined to some angle; whereupon the drop was grown in $10\ \mu\text{l}$ increments, at $0.2\ \mu\text{l/s}$, until the drop was observed to slide while water was *not* being added to the drop. Again, the sliding angle was defined by a discernible movement within a 5 s pause, which began after each $10\ \mu\text{l}$ addition of water was completed.

Care was taken to ensure that the needle tip was never located near the edge of the drop because the drop would then cling to the needle, resulting in a much higher sliding angle. With the

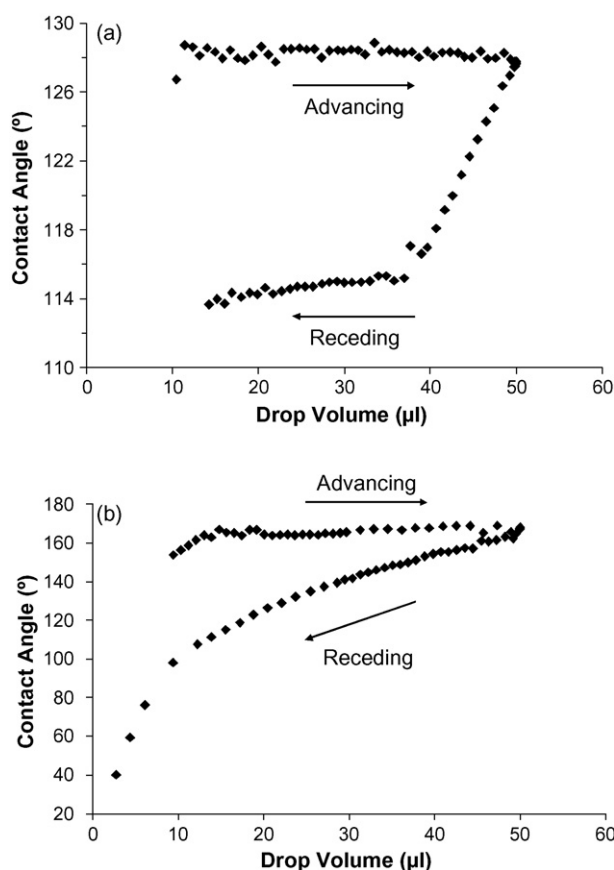


Fig. 4. (a) Typical advancing and receding contact angle data for water drop on fluorinated silicon surface. (b) Typical advancing and receding contact angle data for water drop on AKD surface. Water was added and withdrawn from the drop while the surface remained level.

needle tip in the centre of the drop, adhesion forces between the needle and the drop were observed to be much lower but were still anticipated to have some effect on the sliding angle. Therefore, a conventional method (drop placed on a level surface, then inclined) benchmark was taken with the needle likewise immersed in the drop. Thus, the effect of placing a drop on an inclined surface (rather than the conventional method) could be investigated without concern for any adhesion between the needle and the drop. Again, experiments were repeated between three and four times for each data point.

4. Results and discussion

4.1. Advancing and receding contact angle measurement

As expected, the advancing and receding contact angles of water on both AKD and fluorinated silicon surfaces were unaffected by the maximum drop volume used (i.e. 30, 50, or 70 μl). The 50 μl contact angle curves for AKD and silicon (Fig. 4) are representative of these measurements. The advancing and receding contact angles for silicon are $128 \pm 0.2^\circ$ and $115 \pm 0.2^\circ$, respectively. The advancing contact angle for AKD is $165 \pm 2^\circ$, but the receding angle is too small to be measured accurately with ADSA; by observation, it

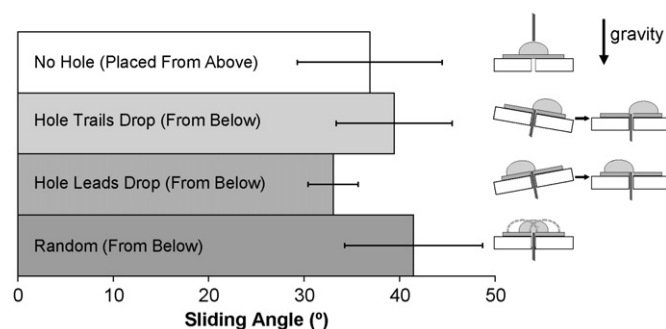


Fig. 5. Water drops were placed on level AKD surface samples using four different methods and then inclined. The graph shows the mean sliding angles for each placement method, with one standard deviation error. A *t*-test (5% alpha level) concluded that only the “hole leads drop” results were statistically distinct from the others.

appeared to be less than 10° (pinning of contact line was observed).

4.2. Drop placement in conventional tilted plate experiments

The sliding angles with one standard deviation error for each of the four water drop placement methods on AKD surfaces are presented in Fig. 5. These measurements were all taken using the conventional tilted plate technique: drops placed on a level surface, then inclined. A *t*-test with a 5% alpha level confirmed that the only placement method that gives statistically distinct results from those of the other techniques is the “hole leading drop” method. Although the statistical differences between these methods are small, it is important to understand their cause and possible relevance to drop mobility studies.

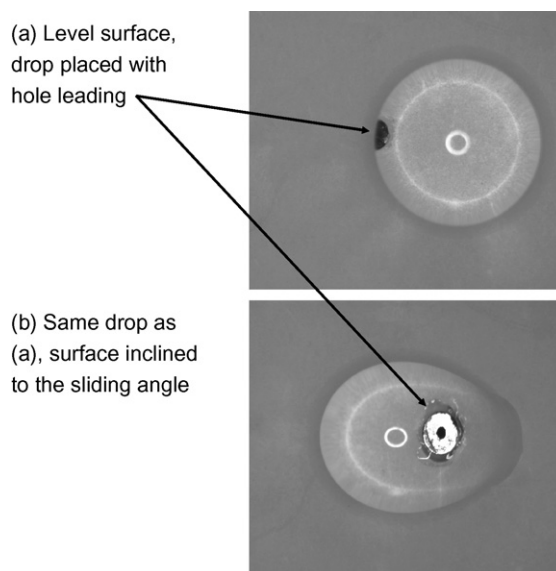


Fig. 6. Top view of water drop on AKD in “leading hole” position before (a) and after (b) inclining: contact line does not pin on the hole at incipient sliding. The end of the needle and hole can be clearly seen in (b), but appear only as a dark spot near the edge of the drop in (a). The gravity vector points into the page in (a).

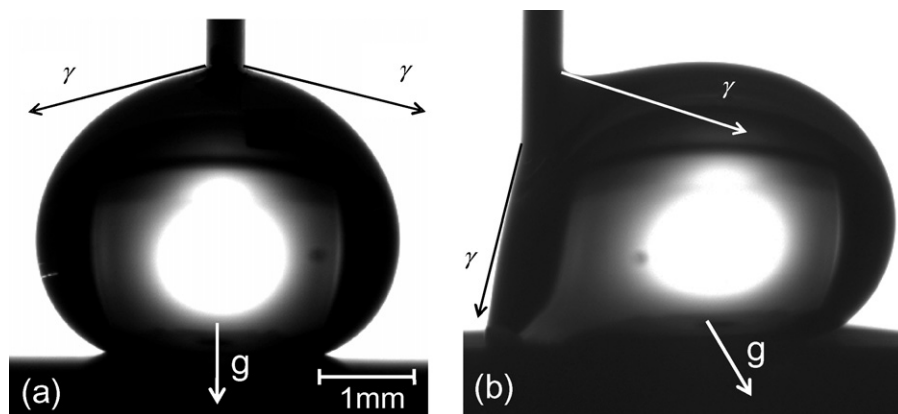


Fig. 7. (a) When the needle tip is in the center of the drop, surface tension forces around the periphery of the needle oppose each other, leaving a small resultant reaction force. (b) When the needle tip nears the edge of the drop, the directions of the surface tension forces change, and the resultant reaction force increases. Note that the drop is inclined in (b) but not in (a).

Drops placed with the hole leading drop technique resulted in lower sliding angles than other placement methods from below. The reason is that the trailing edge of the drop did not tend to pin on the hole before sliding. As the drop was inclined, its leading edge would move forward until the contact line did not lie on the hole (Fig. 6). The contact line would remain in this position until the surface was inclined to the sliding angle, when the drop began to slide.

The leading hole method also resulted in a mean sliding angle measurement that was 24% (8°) lower than that of placement from above. Drops placed from above had water added to them from a needle that was positioned just above their surface. As water was added, the drops would visibly vibrate and as a result, often displayed less than an advancing contact angle after placement. This situation would result in a slightly larger wetted area (and contact line length) for a given drop volume, and subsequently, higher sliding angles. The random vibration caused by placing drops from above also led to somewhat inconsistent sliding angle measurements. Among those drops placed from below, measurement precision was affected by the pinning of drops on holes, which had varying edge conditions. For these reasons, the standard deviation of measurements using the leading hole method was 50–60% ($3\text{--}5^\circ$) less than the other placement techniques.

Drops placed with the hole leading were observed to interact least with the hole while also avoiding the detrimental effects of vibration caused by placement from above. Therefore, hole leading drop placement is recommended whenever possible in tilted plate experiments using hydrophobic surfaces.

4.3. Effect of needle immersed in drop on sliding angle measurements

In researching some physical problems, it may be useful to add or withdraw liquid from a drop while it is resting on an inclined surface. One such application could be fluid transfer through an inclined membrane, which subsequently slides off as droplets (e.g. fuel cell); a more common example might be the condensation of drops on inclined surfaces. Section 3.4 described a technique used in this study that involved inject-

ing water through a needle that was immersed into a drop on an inclined surface. The needle tip was placed as near as possible to the center of the drop, where the resultant reaction force between the needle and the sliding drop was observed to be minimized (Fig. 7).

Fig. 8 shows that the difference between sliding angle measurements with and without needle immersion is always less than

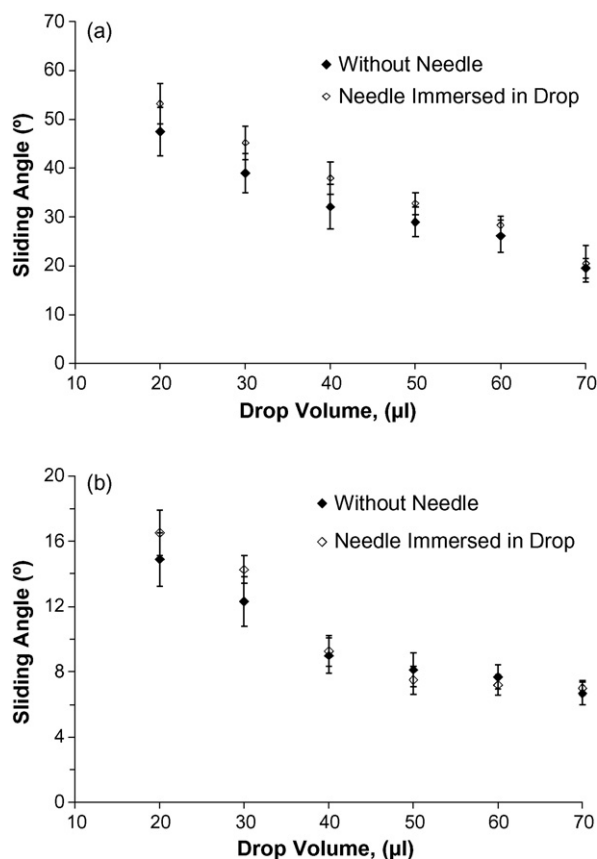


Fig. 8. For water on AKD (a) and fluorinated silicon surfaces (b), effect of needle immersed in drop is less than that of the combined uncertainty of the measurements (one standard deviation). For AKD, the observed effect is more pronounced at lower drop volumes. For fluorinated silicon, needle immersion does not affect the sliding angles of drops larger than $40\text{ }\mu\text{l}$.

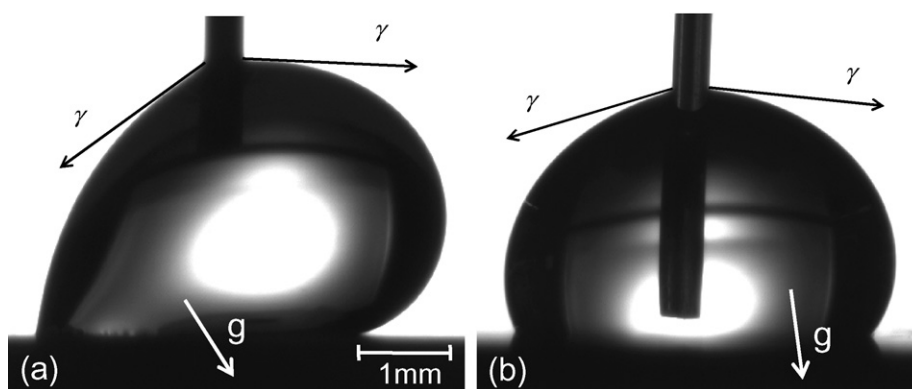


Fig. 9. Drops about to slide on AKD surface (a) are more elongated than those on fluorinated silicon (b). The dissimilar directions of the surface tension forces in (a) lead to a higher resultant reaction force between the drop and the needle. Note that the camera is inclined with the experimental stage, so the surface appears to be horizontal in these images.

their combined uncertainty of one standard deviation. The effect of needle immersion appears to be greatest at lower drop volumes (less than 70 μl for AKD and 40 μl for fluorinated silicon), likely because the resultant reaction force between the needle tip and drop is insignificant compared to the adhesion force required to retain large drops on the surface. Drops on AKD surfaces tended to adhere to the needle tip more than those on fluorinated silicon surfaces because of differences in drop profile shape. In Fig. 9, it can be seen that the elongated shape of a drop that is about to slide on an AKD surface incurs a higher resultant reaction force than the more symmetric shape of a drop on fluorinated silicon.

For tilted plate experiments with larger drops, especially those with little drop elongation (such as with low contact angle hysteresis surface/liquid systems), immersion of the needle tip in the drop apex has a negligible effect on sliding angle measurements. With proper attention to needle placement, this technique is a valid method for studying drop growth on inclined surfaces of low contact angle hysteresis.

4.4. Drop placement on fixed, inclined surfaces

The results of the Callaghan and Amirfazli [5] experiments with water on AKD and fluorinated silicon are presented in Fig. 10. Fig. 10a illustrates the original observation: that drops placed on inclined AKD surfaces have lower sliding angles than those placed on level surfaces, then inclined. On fluorinated silicon, Fig. 10b, there is no statistically significant difference (based on *t*-test, 5% alpha level) between the two methods, but there appears to be a similar trend as with AKD surfaces. As mentioned in the discussion by Callaghan and Amirfazli, it is unclear if this trend is simply due to kinetic energy input (e.g. hand vibration, etc.) when the drop is placed with a handheld pipette.

Automated drop placement data (Fig. 11) shows a similar, but stronger distinction between the sliding angles of drops placed on inclined surfaces and drops placed on level surfaces, then inclined. Since drop growth using the automated placement technique was very slow and free of operator induced vibrations, these findings reveal that the behavior observed by Callaghan and Amirfazli [5] was not simply due to inadvertent kinetic energy input. There are two factors to consider when comparing

handheld with automated method data. Recall that sliding angles found with the automated method should be slightly lower than with the handheld method because of a necessary change in the way the sliding angle was determined. However, this effect is negated by retention forces between the needle tip and the drop (only present with automated method), especially for AKD surfaces and lower drop volumes.

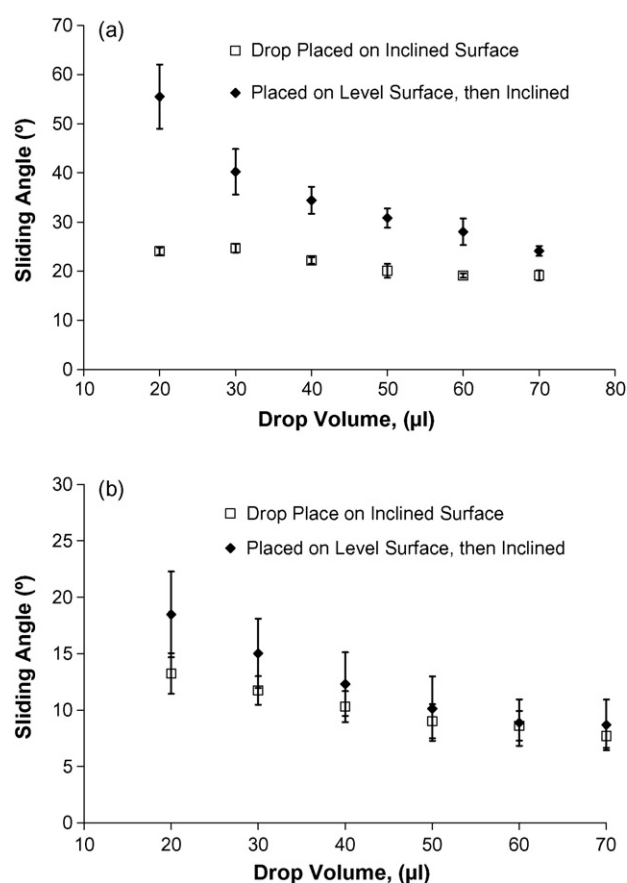


Fig. 10. (a) Results for handheld drop placement on AKD surface show a significant difference between the two methods at lower volumes, but the difference diminishes at higher volumes. (b) For handheld drop placement on fluorinated silicon, a *t*-test (5% alpha level) reveals no statistically significant difference between the two methods, but there appears to be a similar trend as with AKD surfaces.

Table 1a

Summary of contact angle data for tilted plate experiments with 30 μl drops

	AKD			Fluorinated silicon		
	$\theta_{\max}$ ($^{\circ} \pm \text{S.D.}$)	$\theta_{\min}$ ($^{\circ} \pm \text{S.D.}$)	$\theta_{\max} - \theta_{\min}$ ($^{\circ} \pm \text{S.D.}$)	$\theta_{\max}$ ($^{\circ} \pm \text{S.D.}$)	$\theta_{\min}$ ($^{\circ} \pm \text{S.D.}$)	$\theta_{\max} - \theta_{\min}$ ($^{\circ} \pm \text{S.D.}$)
Tilted plate method						
Drop placed on inclined surface	161 $\pm$ 5	86 $\pm$ 3	75 $\pm$ 8	126 $\pm$ 3	113 $\pm$ 4	13 $\pm$ 7
Conventional (level surface, then inclined)	163 $\pm$ 5	83 $\pm$ 6	80 $\pm$ 11	124 $\pm$ 4	112 $\pm$ 3	12 $\pm$ 7

This data clearly illustrates that conventional tilted plate experiments (drop placed on level surface, then inclined) cannot be assumed to adequately represent the physical problem of drops growing and shedding from an inclined surface, such as in a condenser. For both a high (AKD) and a low (fluorinated silicon) contact angle hysteresis surface/liquid combination, the error of this assumption would be as high as 50–60%. In understanding drop mobility on a fixed, inclined surface, the conventional tilted plate method would not be appropriate.

The reason for this deviation lies in the shape of drops produced by either method. Despite the observed difference in sliding angles, the maximum and minimum contact angles for water on both surfaces were not affected by drop placement

method (Table 1a). However, top views of drops at the point of sliding on inclined AKD exhibit a distinct difference in contact line shape. Being a high contact angle hysteresis system, a water drop that has been placed on a level AKD surface, then inclined, tends to pin, leading to a more elongated contact line with a wide trailing edge, shown in Fig. 12a. Recall that drop adhesion is the result of the component of the surface tension force distribution (Fig. 1b) that is perpendicular to the axis of inclination. The shape in Fig. 12a is more effective in adhering to the surface than the near teardrop shape that forms when a drop is grown on the inclined surface (Fig. 12b). Although it was only possible to observe the maximum and minimum contact angles of the drop, it is intuitive that changes in contact line shape would be accompanied by changes in intermediate contact angles. It is important to note that the observed differences in drop adhesion are the combined result of changes in the contact line shape and subsequent intermediate contact angle distribution. Calculations of k -value (the fitting parameter used in application of Eqs. (1) or (2)), which is meant to capture the combined effect of contact line shape and contact angle distribution (along the contact line) confirms this. Using Eq. (2) and results from Table 1a, and Figs. 10a and 11a, the k values were found as approximately 1.8 and 1.15, for conventional and pre-inclined surfaces, respectively (the k values are representative of both manual and automated drop placement procedures). Interestingly, if Eq. (1) was used to calculate the k -value, an unreasonable negative value for k would have resulted (note that in this case data from Table 1b should be used). This is another indication that Eq. (2) is the appropriate equation to use for interpretation of data, rather than Eq. (1) as discussed earlier; this is regardless of the protocol used for the experimentation.

Although the relative difference in sliding angles for fluorinated silicon surfaces was similar to that on AKD surfaces (50–60%), the top views of drops on fluorinated silicon do not show such an obvious difference in shape (Fig. 13). Therefore, a very small change in adhesion force for fluorinated silicon produced the same relative difference in sliding angle as a much larger change for AKD. The explanation lies in the respective magnitudes of these sliding angles, as the adhesion force is not proportional to the sliding angle, but its cosine (see Fig. 1a). To illustrate, consider the case of a 30 μl drop: the sliding angles for AKD were 22 $^{\circ}$ for drops placed on an inclined surface and 45 $^{\circ}$ for the conventional method (51% difference), and for fluorinated silicon, they were 6 $^{\circ}$ and 14 $^{\circ}$ (57% difference). However, the corresponding difference in adhesion forces is 25% for AKD and only 4% for fluorinated silicon. Although the shift in contact

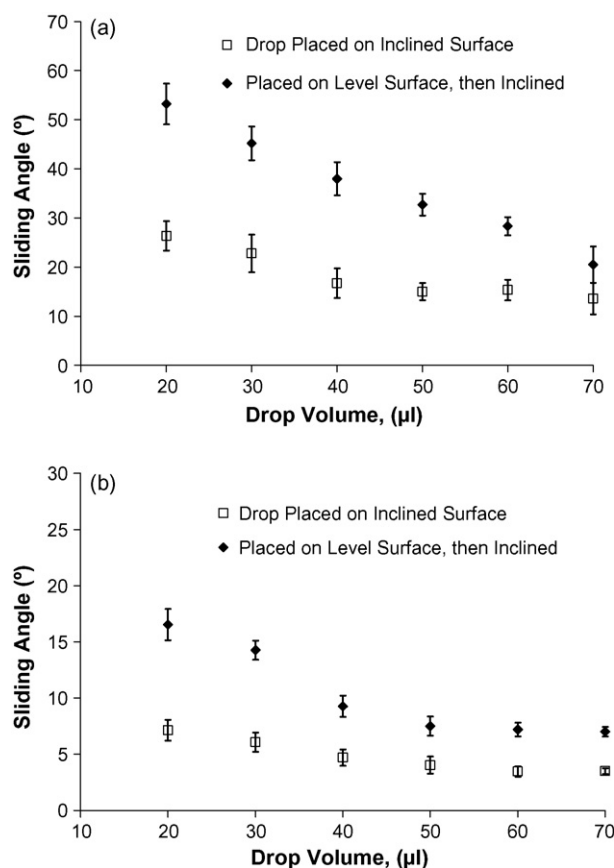


Fig. 11. (a) Automated drop placement results show that water drops placed on inclined AKD surfaces have significantly lower sliding angles, especially at lower drop volumes. (b) Automated results for water on fluorinated silicon surface show a significant difference between the two methods as well.

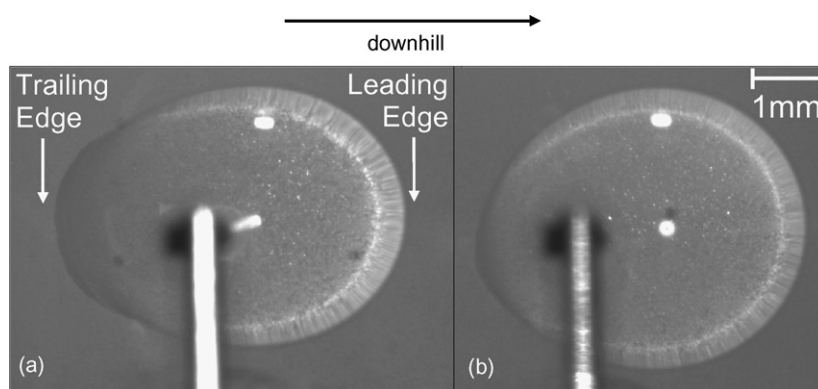


Fig. 12. Top view of 30 μl drops on inclined AKD surface: (a) placed on level, then inclined to 48°, (b) placed on 22° incline. Both drops shown are at their respective sliding angles. The bright rectangular areas are reflections from the needle.

Table 1b

Summary of contact angle data for drop expansion/contraction on level surface

AKD			Fluorinated silicon		
θ_A (° $\pm$ S.D.)	θ_R (°)	CA hysteresis (°)	θ_A (° $\pm$ S.D.)	θ_R (° $\pm$ S.D.)	CA hysteresis (° $\pm$ S.D.)
165 $\pm$ 2	<10	>155	128 $\pm$ 0.2	115 $\pm$ 0.2	13 $\pm$ 0.4

line shape for fluorinated silicon is less pronounced than in the case of AKD, it must be assumed that it exists. This is confirmed by the different k -value found for each system. The k values for fluorinated silicon surface were found to be approximately 5 and 3 for conventional and pre-inclined surface experiments, respectively.

Comparing the data in Tables 1a and 1b, there is no significant difference between the advancing/receding and maximum/minimum contact angles for fluorinated silicon surfaces. This result is in accordance with the findings of ElSherbini and Jacobi [4]: that θ_A and θ_R are equal to $\theta_{\max}$ and $\theta_{\min}$. However, the receding contact angle of water on the AKD surfaces is at least 73° less than the minimum angle. This behavior does not agree with the theoretical study by Krasovitz and Marmur [3], which predicted that a receding contact angle less than the minimum angle is only expected from a hydrophilic surface, defined by the authors as having an average contact angle $((\theta_A + \theta_R)/2)$ of

30°. It also obviously does not agree with ElSherbini and Jacobi [4], possibly because their conclusions were based on cases where θ_A was less than 112° and greater than 49°. In systems of higher repellency, especially those with textured surfaces, there may be wetting mechanisms which are not represented in the data set from [4]. Considering the prominence of ultra-hydrophobic surface research in the recent literature, it would be prudent to recognize the possibility of other surfaces that behave like AKD, and draw no indiscriminate parallels between the advancing/receding and maximum/minimum contact angles.

The results of this study also illustrate the danger of associating repellency with drop mobility. From the advancing contact angle data in Table 1b, it is clear that AKD surfaces are more water repellent than fluorinated silicon surfaces. However, the tilted plate data in this study shows that the sliding angles of water drops on AKD surfaces are at least double those for fluorinated silicon surfaces. This apparent paradox is explained by the

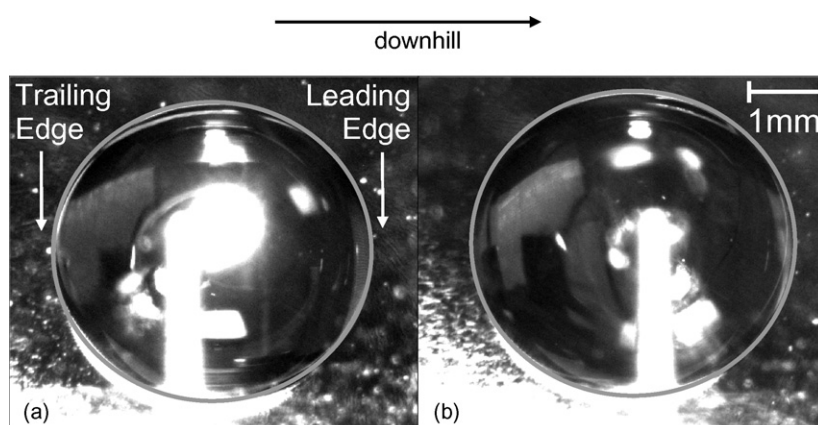


Fig. 13. Top view of 30 μl drops (outline is highlighted in grey for graphical clarity) on inclined fluorinated silicon surface: (a) placed on level, then inclined to 16°, (b) placed on 7° incline. Both drops shown are at their respective sliding angles. The bright rectangular areas are reflections from the needle.

difference in inclined contact angles ($\theta_{\max} - \theta_{\min}$) presented in Table 1a. The difference between $\theta_{\max}$ and $\theta_{\min}$ is much greater for AKD than for fluorinated silicon, resulting in a higher sliding angle, as predicted by Eq. (2).

5. Conclusions

The methodology of tilted plate experiments has been investigated with respect to sliding angle measurements. A novel drop placement technique that emulates the condensation of a liquid on an inclined surface has also been introduced, leading to the confirmation and explanation of results from Callaghan and Amirfazli [5]. The detailed discussion in this investigation has also served to bridge a gap in the documentation of sliding angle measurement protocol.

Associations between the advancing/receding and maximum/minimum contact angles must be approached with caution. In this study, it was found that the receding contact angle was much less (74° difference) than the minimum contact angle for water drops on AKD. This result agrees with neither the theoretical predictions from Krasovitski and Marmur [3] nor the empirical predictions of ElSherbini and Jacobi [4]. Those of the latter were based on data for systems with θ_A between 49° and 112° . It was suggested that special circumstances may arise for superhydrophobic systems like AKD and water ($\theta_A = 165^\circ$). Conversely, observations of water on fluorinated silicon surface ($\theta_A = 128^\circ$) agreed with the predictions of ElSherbini and Jacobi, exhibiting almost identical maximum/minimum and advancing/receding angles.

In placing drops for conventional tilted plate experiments (placed on level surface, then inclined), there are two major problems: drop spreading due to vibration when placed from above and pinning on the needle hole when placed from below. Experiments with water drops on AKD surface showed that these problems had a statistically significant affect (*t*-test, 5% alpha level) on the magnitude of sliding angle measurements. They were minimized with the “hole leading drop” placement, which resulted in measurements that were more than twice as precise as those of the other placements tested. This drop placement technique is recommended whenever possible.

The growth and shedding of drops on an inclined surface, such as in a condenser, was emulated on AKD and fluorinated silicon surfaces. It was shown that drops placed on inclined surfaces

have lower sliding angles than those placed conventionally (on a level surface, then inclined). However, the maximum and minimum contact angles for both surfaces were unchanged between the two experimental methods, so the change in adhesion had to be a result of different contact line shapes. This difference was visually verified with top views of the drops on AKD. However, the contact line shape variation for fluorinated silicon was too small to be visually distinguishable.

Can we predict mobility behavior for an application such as a condenser based on the results of conventional tilted plate experiments or the advancing contact angle (repellency)? In the case of both a high and low contact angle hysteresis system, it has been shown that the conventional measurement would have been in error by 50–60%. Furthermore, it was shown that advancing/receding contact angle measurements (on a level surface) cannot be relied upon to predict drop mobility. The repellency of the test surfaces was completely misleading in terms of their relative drop mobility, and the advancing/receding contact angles did not consistently correspond to the maximum/minimum contact angles (on a tilted surface). These results clearly underline the importance of tilted plate experiment design and interpretation.

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